

Exp No: 8- GAS SENSOR MQ-2 IN PROTEUS USING ARDUINO

Objectives:

1. To interface the MQ-2 gas sensor with the Arduino UNO.
2. To detect gas leakage using the MQ-2 sensor.
3. To understand gas sensor interfacing and monitoring using Arduino in Proteus.

Program:

```
#define GreenLed 13
```

```
#define Sensor A0
```

```
void setup(){
```

```
pinMode(13,OUTPUT);
```

```
pinMode(12,OUTPUT);
```

```
pinMode(A0,INPUT);
```

```
Serial.begin(9600);
```

```
}
```

```
void loop() {
```

```
int value = analogRead(A0);
```

```
Serial.print("Analogic Value coming from the sensor : ");
```

```
Serial.println(value);
```

```
delay(100);
```

```
if(value>600){
```

```
digitalWrite(13,HIGH);
```

```
}
```

```
else{
```

```
digitalWrite(13,LOW);
```

```
}
```

```
delay(20);
```

```
}
```

Code for MQ6 sensor:

```
int mq6Pin = A0; // MQ-5 analog pin connected to A0
```

```
void setup() {
```

```
Serial.begin(9600);
```

```
}
```

```
void loop() {
```

```
int sensorValue = analogRead(mq5Pin);
```

```
Serial.print("MQ5 Sensor Value: ");
```

```
Serial.println(sensorValue);
```

```
delay(1000); // 1-second delay
```

```
}
```

Output:

